

NAIVE BAYES CLASSIFICATION

- NOTES FROM READING 'NAIVE BAYES AND TEXT CLASSIFICATION' - SEMSTIN RASCHKA + OTHER RESOURCES.
- SUB-FIELD OF PREDICTIVE MODELLING → SUPERVISED PATTERN CLASSIFICATION
 - ↳ TRAIN MODEL USING LABELED TRAINING DATA TO ASSIGN PRE-DEFINED CLASS LABEL TO NEW OBJECT.
- NAIVE BAYES CLASSIFIERS → BASED ON BAYES' PROBABILITY THEOREM. → GOOD FOR DOCUMENT CLASSIFICATION + DISEASE PREDICTION.
- LINEAR CLASSIFIERS → SIMPLE YET EFFICIENT
- 'NAIVE' → ASSUMPTION THAT FEATURES IN DATASET ARE MUTUALLY INDEPENDENT → IN PRACTICE, OFTEN VIOLATED → STILL PERFORMS UNDER UNREALISTIC ASSUMPTIONS.
- SMALL SAMPLE SIZES → CAN OUTPERFORM MORE POWERFUL ALTERNATIVES
- STRONG VIOLATIONS OF INDEPENDENCE ASSUMPTIONS + NON-LINEAR CLASSIFICATION PROBLEMS → VERY POOR PERFORMANCE

POSTERIOR PROBABILITIES

- BAYES' RULE → THOMAS BAYES, PROBABILITY MODEL:

$$\text{POSTERIOR PROBABILITY} = \frac{\text{CONDITIONAL PROBABILITY} \cdot \text{PRIOR PROBABILITY}}{\text{EVIDENCE}} \quad \text{"WHAT IS THE PROBABILITY THAT OBJECT BELONGS TO CLASS } i \text{"}$$

↓

$$P(w_j | x_i) = \frac{P(x_i | w_j) \cdot P(w_j)}{P(x_i)}$$

- OBJECTIVE IS TO MAXIMISE POSTERIOR PROBABILITY, WITH TRAINING DATA TO FORM DECISION RULE

CLASS-CONDITIONAL PROBABILITIES

- ONE ASSUMPTION NAIVE CLASSIFIERS MAKE → SAMPLES ARE i.i.d. (INDEPENDENT AND IDENTICALLY DISTRIBUTED)
- ANOTHER ASSUMPTION → CONDITIONAL INDEPENDENCE OF FEATURES
 - ↳ UNDER NAIVE ASSUMPTIONS → CLASS-CONDITIONAL PROBABILITIES (LIKELIHOOD) → ESTIMATED DIRECTLY FROM TRAINING DATA.

$$P(x | w_j) = P(x_1 | w_j) \cdot P(x_2 | w_j) \cdot \dots \cdot P(x_d | w_j) = \prod_{k=1}^d P(x_k | w_j)$$

↳ 'HOW LIKELY IS IT TO OBSERVE PARTICULAR PATTERN x GIVEN IT BELONGS TO w_j '

- 'INDIVIDUAL' LIKELIHOOD → ESTIMATED VIA MAXIMUM-LIKELIHOOD (FREQUENCY FOR CATEGORICAL DATA):

$$\hat{P}(x_i | w_j) = \frac{N_{x_i, w_j}}{N_{w_j}}, (i = (1, \dots, d)) \quad \begin{array}{l} N_{x_i, w_j} = \text{NUMBER OF TIMES } x_i \text{ IN } w_j \\ N_{w_j} = \text{TOTAL FEATURES COUNT IN } w_j \end{array}$$

- EXAMPLE: 100 DOCUMENTS, 100 DOCUMENTS ARE SPAM → CALCULATE CLASS-CONDITIONAL PROBABILITY FOR NEW MESSAGE 'HELLO WORLD' GIVEN IT'S SPAM.

$$P(x = [\text{HELLO}, \text{WORLD}] | w = \text{SPAM}) = P(\text{HELLO} | \text{SPAM}) \cdot P(\text{WORLD} | \text{SPAM})$$

MAXIMUM LIKELIHOOD ESTIMATE:

$$\hat{P}(x = [\text{HELLO}, \text{WORLD}] | w = \text{SPAM}) = \frac{20}{100} \cdot \frac{2}{100} = 0.004$$

- RESPECT TO NAIVE ASSUMPTION OF CONDITIONAL INDEPENDENCE, WE HAVE A PROBLEM:
 - NAIVE ASSUMPTION - PARTICULAR WORD DOES NOT INFLUENCE THE CHANCE OF ENCOUNTERING OTHER WORDS IN SAME DOCUMENT

PRIOR PROBABILITIES

- CONTEXT OF PATTERN CLASSIFICATION → PRIOR PROBABILITIES CALLED CLASS PRIORS → THE GENERAL PROBABILITY OF ENCOUNTERING A PARTICULAR CLASS
- CASE OF SPAM CLASSIFICATION:

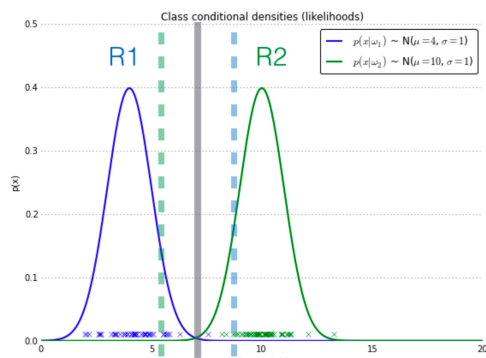
$$P(\text{SPAM}) = \text{"THE PROBABILITY THAT ANY NEW MESSAGE IS A SPAM MESSAGE"}$$

$$P(\text{HAM}) = 1 - P(\text{SPAM})$$

- IF PRIORS ARE FOLLOWING UNIFORM DISTRIBUTION $\rightarrow$ POSTERIOR PROBABILITIES WILL ENTIRELY DETERMINE CLASS-CONDITIONAL PROBABILITIES + EVIDENCE TERM.

$$\hat{P}(w_j) = \frac{N_{w_j}}{N_c} \quad \begin{array}{l} N_{w_j} = \text{SAMPLE COUNT FROM CLASS } w_j \\ N_c = \text{COUNT OF ALL SAMPLES} \end{array}$$

$$\hat{P}(\text{SPAM}) = \frac{\% \text{ OF SPAM MESSAGES IN TRAINING DATA}}{\% \text{ OF ALL MESSAGES IN TRAINING DATA}} \leftarrow \text{CONTEXT OF SPAM CLASSIFICATION}$$



- BELL-CURVE DENOTES THE PROBABILITY DENSITIES OF SAMPLES THAT WERE DRAWN FROM THE TWO DIFFERENT NORMAL DISTRIBUTION.

if $P(w_1) < P(w_2)$ if $P(w_1) = P(w_2)$ if $P(w_1) > P(w_2)$

EFFECT OF PRIOR PROBABILITIES ON DECISION RULE

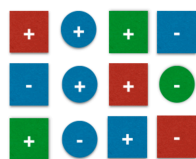
EVIDENCE

- EVIDENCE - $P(x) \rightarrow$ PROBABILITY OF ENCOUNTERING A PARTICULAR PATTERN $\propto$ INDEPENDANT FROM THE CLASS LABEL.

- GIVEN POSTERIOR PROBABILITY IS $\rightarrow P(w_j | x_i) = \frac{P(x_i | w_j) \cdot P(w_j)}{P(x_i)} \rightarrow \text{EVIDENCE} \rightarrow P(x_i) = P(x_i | w_1) \cdot P(w_1) + P(x_i | w_2^c) \cdot P(w_2^c)$ $w_2^c = \text{NOT CLASS } w_2$

MULTINOMIAL NAIVE BAYES

- TOP EXAMPLE



- w_j - CLASS LABEL = $w_j \in \{+, -\}$
 - x_i - 2D FEATURE VECTOR = $x_i = [x_{i1}, x_{i2}]$, $x_{i1} \in \{\text{BLUE, GREEN, RED, YELLOW}\}$, $x_{i2} \in \{\text{CIRCLE, SQUARE}\}$
 - (CAN BE WRITTEN AS:
 $x_i = [x_{i1}, x_{i2}]$
 for $i \in \{1, 2, \dots, n\}$, $n=12$
 and $x_{i1} \in \{\text{BLUE, GREEN, RED, YELLOW}\}$, $x_{i2} \in \{\text{CIRCLE, SQUARE}\}$

- NOW THE TASK IS CLASSIFY NEW SAMPLE - PRETENDING WE DON'T KNOW TRUE LABEL CLASS '+'

MAXIMUM-LIKELIHOOD ESTIMATES

- DECISION RULE:

CLASSIFY SAMPLE AS + IF

$$P(w = + | x = [\text{BLUE, SQUARE}]) \geq P(w = - | x = [\text{BLUE, SQUARE}])$$

ELSE CLASSIFY AS -.

- UNDER ASSUMPTIONS THAT SAMPLES ARE i.i.d $\rightarrow$ PRIOR PROBABILITIES OBTAINED BY MAXIMUM-LIKEHOOD

$\hookrightarrow$ IF SAMPLES ARE NOT i.i.d THEN MLE FOR PRIOR PROBABILITIES WOULD LIKELY BE BIASED + INACCURATE.

$$P(+)=\frac{7}{12}=0.58$$

$$P(-)=\frac{5}{12}=0.42$$

- UNDER 'NAIVE' ASSUMPTION THAT FEATURES 'COLOR' + 'SHAPE' ARE MUTUALLY INDEPENDANT $\rightarrow$ CLASS-CONDITIONAL PROBABILITIES $\rightarrow$ CALCULATED AS DOT PRODUCT.

$$P(x|+) = P(\text{blue}|+) \cdot P(\text{square}|+) = \frac{3}{7} \cdot \frac{2}{7} = 0.31$$

$$P(x|-) = P(\text{blue}|-) \cdot P(\text{square}|-) = \frac{2}{5} \cdot \frac{3}{5} = 0.36$$

- POSTERIOR PROBABILITIES = DOT PRODUCT OF CLASS-CONDITIONAL + PRIOR PROBABILITIES:

$$P(+|x) = P(x|+) \cdot P(+) = 0.31 \cdot 0.58 = 0.18$$

$$P(-|x) = P(x|-) \cdot P(-) = 0.36 \cdot 0.42 = 0.15$$

CLASSIFICATION

- NEW SAMPLES CLASSIFIED BY PLOTTING INTO POSTERIOR PROBABILITIES INTO DECISION RULE:

IF $P(+|x) \geq P(-|x)$ $0.18 > 0.15 \rightarrow$ SAMPLE CLASSIFIED AS +.

CLASSIFY AS +,

ELSE CLASSIFY AS -

ADDITIVE SMOOTHING

- NEW EXAMPLE $\rightarrow$ 'NEW' VALUE $\rightarrow x = [\text{yellow}, \text{square}]$
- IF YELLOW SQUARE $\rightarrow$ NOT IN TRAINING DATASET $\rightarrow$ CLASS-CONDITIONAL + POSTERIOR PROBABILITY = 0
- $\hookrightarrow$ TO AVOID THIS = SMOOTHING TERM ADDED WITH MULTINOMIAL BAYES.
- $\hookrightarrow$ MOST COMMON $\therefore$ LIDSTONE SMOOTHING ($\alpha < 1$)
- LAPLACE SMOOTHING ($\alpha = 1$)

$$\hat{P}(x_i | w_j) = \frac{N_{x_i, w_j} + \alpha}{N_{w_j} + \alpha \cdot d} \quad (i = (1, \dots, d))$$

N_{x_i, w_j} = # TIMES x_i IN w_j
 N_{w_j} = TOTAL FEATURE COUNT OF w_j
 α = ADDITIVE SMOOTHING PARAMETER
 d = DIMENSIONALITY OF FEATURE VECTOR $x = [x_1, \dots, x_d]$

NAIVE BAYES FOR TEXT CLASSIFICATION

- SIMILAR TO INFO IN BOOK $\rightarrow$ CAN LOOK AT THOSE NOTES

BAG-OF-WORDS MODEL

- THREE MAIN CRITERIA OF GOOD FEATURES:
 - SALIENT - FEATURES ARE IMPORTANT + MEANINGFUL WITH RESPECT TO PROBLEM DOMAIN
 - INVARIANT - FEATURES INSUSCEPTIBLE TO DISTORTION, SCALING, ORIENTATION, ETC...
 - DISCRIMINATORY - FEATURES BARE ENOUGH INFO $\rightarrow$ DISTINGUISH WELL BETWEEN PATTERNS.
- BEFORE FITTING MODEL $\rightarrow$ REPRESENT TEXT DOCUMENT AS FEATURE VECTOR
 - $\hookrightarrow$ BAG-OF-WORDS $\rightarrow$ COLLECTION OF DIFFERENT WORDS THAT OCCUR IN TRAINING SET + EACH WORD ASSOCIATED WITH COUNT OF OCCURANCE
- VOCAB USED TO CONSTRUCT d-DIMENSIONAL FEATURE VECTOR FOR EACH DOCUMENT $\rightarrow$ DIMENSIONALITY = NUMBER OF DIFFERENT WORDS IN VOCAB. $\Rightarrow$ VECTORIZATION

TOKENIZATION

- BREAKING DOWN TEXT CORPUS INTO INDIVIDUAL ELEMENTS $\rightarrow$ INPUTS FOR VARIOUS NLP ALGOS.
- USUALLY ACCOMPANIED BY OTHER OPTIONAL PROCESSING STEPS $\rightarrow$ E.G. STOP WORD + PUNCTUATION REMOVAL, STEMMING, LEMMATIZING, CONSTRUCTING n-GRAMS

STOP WORDS

- STOP WORDS $\rightarrow$ COMMON WORDS IN TEXT $\rightarrow$ CONSIDERED UN-INFORMATIVE (E.G. AND, OR, THE, ...)
- ONE APPROACH $\rightarrow$ SEARCH AGAINST LANGUAGE-SPECIFIC STOP WORD DICTIONARY.
- ANOTHER APPROACH $\rightarrow$ CREATE STOP LIST BY SORTING ALL WORDS IN ENTIRE TEXT CORPUS BY FREQUENCY.

STEMMING + LEMMATIZATION

- STEMMING $\rightarrow$ TRANSFORMING A WORD INTO ROOT FORM. $\rightarrow$ MARTIN F. PORTER (1979) $\rightarrow$ PORTER STEMMER
 - $\hookrightarrow$ CAN CREATE NON-REAL WORDS $\rightarrow$ 'THU' INSTEAD OF 'THIS'.
- LEMMATIZATION $\rightarrow$ OBTAIN CANONICAL FORMS OF WORDS, CALLED LEMMAS
 - $\hookrightarrow$ COMPUTATIONALLY MORE DIFFICULT + EXPENSIVE OVER STEMMING

N-GRAMS

- TOKEN DEFINED AS SEQUENCE OF n ITEMS.
- UNIGRAM (1-GRAM), BIGRAM (2-GRAM), TRIGRAM (3-GRAM)
- EACH WORD CONSISTS OF ONE WORD/LETTER/SYMBOL
- n DEPENDS ON LANGUAGE + APPLICATION
 - ↳ ENGLISH TEXT BOOK, $4 \leq n \leq 8$
 - ↳ SERBIAN TEXT DOCUMENT, $3 \leq n \leq 7 \rightarrow$ ANDELKA ZECEVIC
 - ↳ ANTI-SPAM EMAIL FILTERING, $n=3$ AND $4 \rightarrow$ KANARIS e. al.

DECISION RULE FOR SPAM CLASSIFICATION

- DECISION RULE $\rightarrow$ POSTERIOR PROBABILITIES:

$$P(x | w = \text{spam}) \geq P(x | w = \text{ham})$$

- POSTERIOR PROBABILITIES = PRODUCT OF CLASS-CONDITIONAL + PRIOR PROBABILITY:

$$P(x | w = \text{spam}) = P(w = \text{spam} | x) \cdot P(\text{spam})$$

$$P(x | w = \text{ham}) = P(w = \text{ham} | x) \cdot P(\text{ham})$$

- PRIOR PROBABILITIES OBTAINED VIA MAXIMUM-LIKELIHOOD ESTIMATES:

$$\hat{P}(w = \text{spam}) = \frac{\% \text{ OF SPAM MSG.}}{\% \text{ OF ALL MSG.}}$$

$$\hat{P}(w = \text{ham}) = \frac{\% \text{ OF HAM MSG.}}{\% \text{ OF ALL MSG.}}$$

- ASSUMING WORDS IN EACH DOCUMENT ARE CONDITIONALLY INDEPENDANT $\rightarrow$ TWO MODELS TO COMPUTE CLASS-CONDITIONAL PROBABILITIES:

↳ MULTI-VARIATE BERNOULLI MODEL

↳ MULTINOMIAL MODEL

MULTI-VARIATE BERNOULLI NAIVE BAYES

- BASED ON BINARY DATA $\rightarrow$ EVERY TOKEN IN FEATURE VECTOR IS EITHER 1 OR 0.
- FEATURE VECTOR HAS m DIMENSIONS $\rightarrow m$ = NUMBER OF WORDS IN WHOLE VOCAB.
- 1 MEANS WORD OCCURS IN DOC, 0 DOES NOT.
- BERNOULLI WRITTEN AS:

$$P(x | w_j) = \prod_{i=1}^m P(x_i | w_j)^{x_i} \cdot (1 - P(x_i | w_j))^{(1-x_i)} \quad (b \in 0, 1)$$

$$\hat{P}(x_i | w_j) = \text{MAXIMUM-LIKELIHOOD} \rightarrow \text{ESTIMATE PARTICULAR WORD (OR TOKEN) } x_i \text{ OCCURS IN } w_j:$$

$$\hat{P}(x_i | w_j) = \frac{df_{x_i, y} + 1}{df_y + 2}$$

$df_{x_i, y}$ = % OF DOCUMENTS IN TRAINING DATASET THAT CONTAIN x_i IN w_j
 df_y = % OF DOCUMENTS IN TRAINING DATASET $\rightarrow$ BELONGS TO w_j
+1 AND +2 = LAPLACE SMOOTHING PARAMETERS

MULTINOMIAL BAYES

TERM FREQUENCY

- ALT APPROACH TO CHARACTERIZE TEXT DOCUMENTS.
- TERM FREQUENCY ($tf(t, d)$) $\rightarrow$ DEFINED AS NUMBER OF TIMES t (WORD/TOKEN) IN d (DOCUMENT) $\rightarrow$ SOMETIMES CALLED RAW FREQUENCY
 - ↳ NORMALIZED BY $\rightarrow$ RAW TERM FREQUENCY / DOCUMENT LENGTH

$$\text{NORMALIZED TERM FREQUENCY} = \frac{tf(t, d)}{n_d} \quad tf(t, d) = \text{RAW TERM FREQUENCY (COUNT } t \text{ IN } d)$$

n_d = TOTAL % OF TERMS IN DOCUMENT d .

- TERM FREQUENCY THEN USED TO COMPUTE MAXIMUM-LIKEHOOD:

$$\hat{P}(x_i | w_j) = \frac{\sum_{d \in w_j} \text{tf}(x_i, d \in w_j) + \alpha}{\sum_{d \in w_j} 1 + \alpha \cdot V}$$

x_i = WORD FROM FEATURE VECTOR x OF PARTICULAR SAMPLE

$\sum \text{tf}(x_i, d \in w_j)$ = SUM OF RAW TERM FREQUENCIES OF WORD x_i , FROM ALL DOCUMENTS IN TRAINING SAMPLE CLASS w_j

$\sum_{d \in w_j} 1$ = SUM OF ALL TERM FREQUENCIES IN TRAINING DATASET FOR CLASS w_j

α = ADDITIVE SMOOTHING PARAMETER

V = SIZE OF VOCAB

- CLASS-CONDITIONAL PROBABILITY OF ENCOUNTERING TEXT x → CALCULATED AS PRODUCT FROM LIKELIHOODS OF INDIVIDUAL WORDS:

$$P(x | w_j) = P(x_1 | w_j) \cdot P(x_2 | w_j) \cdot \dots \cdot P(x_n | w_j) = \prod_{i=1}^n P(x_i | w_j)$$

TERM FREQUENCY - INVERSE DOCUMENT FREQUENCY (TF-IDF)

- ANOTHER ALTERNATIVE FOR CHARACTERISING TEXT DOCUMENTS
- CAN BE UNDERSTOOD AS WEIGHTED TERM FREQUENCY → USEFUL FOR IF STOP WORDS HAVE NOT BEEN REMOVED.
- TF-IDF ASSUMES → IMPORTANCE OF A WORD IS INVERSELY PROPORTIONAL TO HOW OFTEN IT OCCURS ACROSS ALL DOCUMENTS.
- TF-IDF → MOST COMMONLY USED TO RANK DOCUMENTS BY RELEVANCE IN DIFFERENT TEXT MINING TASKS → E.G. PAGE RANKING FOR SEARCH ENGINE
- ALSO APPLICABLE TO TEXT CLASSIFICATION VIA NAIVE BAYES:

$$\text{TF-IDF} = \text{tf}_t(t, d) \cdot \text{idf}(t)$$

- TF-IDF - NORMALIZED TERM FREQUENCY, IDF - INVERSE DOCUMENT FREQUENCY:

$$\text{idf}(t) = \log \left(\frac{N_d}{n_d(t)} \right) \quad \begin{array}{l} N_d = \text{TOTAL \# OF DOCUMENTS} \\ n_d(t) = \# \text{ OF DOCUMENTS THAT CONTAIN TERM } t \end{array}$$

PERFORMANCES OF MULTI-VARIATE BERNOLLI AND MULTINOMIAL MODEL

- MULTINOMIAL > MULTI-VARIATE BERNOLLI - IF VOCAB SIZE IS RELATIVELY LARGE
- PERFORMANCE OF ML ALGO DEPENDANT ON FEATURE CHOICE → MOST IMPORTANT

NAIVE BAYES VARIANTS

GAUSSIAN NAIVE BAYES

- NAIVE BAYES CAN ALSO BE USED ON CONTINUOUS DATA
- IRL FLOWER DATASET → GOOD EXAMPLE OF CONTINUOUS FEATURES
 - ↳ STRATEGY → DISCRETIZE FEATURES + FORM DISTINCT CATEGORIES OR USE GAUSSIAN KERNEL
- GAUSSIAN NAIVE BAYES (ASSUMPTION → PROBABILITY DISTRIBUTIONS OF FEATURES FOLLOW NORMAL (GAUSSIAN DISTRIBUTION)):

$$P(x_i | w) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp \left(-\frac{(x_{ik} - \mu_w)^2}{2\sigma_w^2} \right) \quad \begin{array}{l} \mu = \text{SAMPLE MEAN} \\ \sigma = \text{STANDARD DEVIATION} \end{array}$$

- NAIVE BAYES ASSUMPTION OF CONDITIONAL INDEPENDENCE → CLASS-CONDITIONAL PROBABILITIES COMPUTED AS PRODUCT OF INDIVIDUAL PROBABILITIES:

$$P(x_i | w) = \prod_{k=1}^d P(x_{ik} | w)$$

EAGER + LAZY LEARNING ALGORITHMS

- **EAGER LEARNERS** → FAST WITH CLASSIFYING NEW INSTANCES.
 - ↳ LEARN A MODEL FROM TRAINING DATASET AS SOON AS DATA IS AVAILABLE.
 - ↳ ONCE MODEL LEARNED → NO RE-EVALUATION NEEDED FOR TRAINING DATA TO MAKE NEW PREDICTIONS.
 - ↳ COMPUTATIONALLY MOST EXPENSIVE → MODEL BUILDING STEP
- **LAZY LEARNER** → MEMORISE + RE-EVALUATE TRAINING DATASET FOR PREDICTING CLASS LABELS
 - ↳ ADVANTAGE → MODEL BUILDING (TRAINING) IS FAST
 - ↳ BUT PREDICTION IS SLOW → BECAUSE OF RE-EVALUATION
 - ↳ DISADVANTAGE → TRAINING DATA RETAINED → EXPENSIVE FOR STORAGE
 - ↳ EXAMPLE OF LAZY LEARNER → K-NN